

Baslan Asaad

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EDUCATION

Rutgers University Honors College

Piscataway, NJ

Electrical and Computer Engineering & Computer Science | Expected Graduation: May 2028 | GPA: 3.96/4.00

SKILLS

Software: Python, OpenCV, PyTorch, Linux, GitLab, ROS, Java, C++, KiCad, LTspice, SolidWorks

Robotics: Autonomy, Control Loops, PID, Computer Vision, SLAM, Sensor Fusion, Multithreading

Hardware: Circuit Design, PCB Design, Filters, Arduino, Raspberry Pi, Serial Communications

EXPERIENCE

Columbia University CS3 Accelerator

New York, NY

Technical Lead

Oct. 2026 – Present

- Developing an autonomous routing system to find the lowest-risk paths for commercial fleets and public users, unlocking dynamic insurance data that reduces premiums and redefines road liability.
- Leading customer discovery across 20+ key industry stakeholders to refine technical product requirements and develop a scalable business model.

General Autonomy

Piscataway, NJ

Mechatronics Engineering Intern

Sep. 2025 – Present

- Developing control software for autonomous test vehicles to execute high-speed transient drift maneuvers under dynamic conditions.
- Designing and building electromechanical systems for a robotic drive-by-wire module that remotely actuates steering, throttle, and braking.
- Designed and implemented a lidar-embedded Taxi Sign payload to enable LIO-SAM 3D mapping in dense urban environments.

EV-Edison

Kearny, NJ

Electrical & Software Engineering Intern

May 2026 – Aug. 2026

- Architected and developed EV-Edison's official website with targeted SEO architecture, dynamic location-clustering maps, video-free performant hero animations, and automated contact pipelines via Resend API.
- Designed contractor-approved electrical infrastructure layouts for 6 EV sites under construction, optimizing transformer and charger placement to minimize trenching and validating spatial clearance via vehicle tracking software.
- Developed an internal software tool to automate the EV site design process, accelerating layout generation and electrical placement workflows.

Falcon Attack and Defense Autonomy

Little Falls, NJ

Aerospace & Software Engineering Intern

May 2025 – Aug. 2025

- Developed GPS-denied proportional navigation from scratch for high-speed drones, using camera-based computer vision to achieve reliable kinetic interception of moving targets.
- Built comprehensive debug software tools, including radar visualization, video-stream banks, and 3D real-time flight path reconstruction.
- Developed 2 synthetic data generators and full-stack web application features to streamline the simulation, training, and validation of target detection Markov models.
- Managed GitLab version control workflows, reviewing merge requests across 5 team members and guiding interns through code integration.

PROJECTS (see baslan.dev for more)

GenAI Product Classifier

- Built an AI-powered product classification system using embeddings and LLMs for hierarchical taxonomy prediction, and optimized semantic search using transformer-based models to improve scalability.